DS605 · Fundamentals of Machine Learning

Lecture 01 Notes — From Data to Decisions

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Notes prepared from lecture 01 slides

These notes follow the slide deck in order and expand every example, demo, and discussion prompt into full text, so they can be read and studied independently of the slides.

Opening Hook: A Decision, Not a Dataset

It's 9 a.m., and you run **ShopKart**, an Indian online retailer. Last night, **4,812 customers** placed an order. This morning, your fraud team can only manually review **50** of them. Which 50 do you look at?

The lecture poses four candidate strategies and their consequences:

- **The 50 largest orders.** Reasonable — but fraud is not the same as size. You would spend the morning reading legitimate bulk orders from loyal wholesale buyers.
- **The 50 newest customers.** Sensible instinct, but most new customers are simply new — you would drown in false alarms and annoy the people you are trying to win over.
- **The 50 cash-on-delivery orders.** Closer — CoD does carry more risk — but it is a blunt rule: you would still review thousands of honest CoD orders and miss card fraud entirely.
- **The 50 that look least like normal behavior.** This is the data-mining answer: not a rule someone invented, but a ranking learned from what normal behavior has actually looked like across 2.4 million past orders.

Notice what just happened: the hard part was never the data. It was the **decision** — and the fact that you had to make it with far less certainty than you'd like.

What This Course Is Actually About

Data mining is *not* about finding patterns. It is about **supporting better decisions** when certainty isn't available. Three questions frame every topic in the course:

- **1 · Why does the problem exist?** Something valuable is hidden, and no human can look at 4,812 orders before lunch.
- **2 · How does data mining help?** It narrows an impossible search into a ranked, reviewable shortlist.
- **3 · What decision becomes possible?** Review these 50. Ship the rest tonight. Measure whether you were right.

Every example in the course returns to question 3. If a pattern changes no decision, it is trivia — however clever the algorithm that found it.

Course Logistics

- **Syllabus & evaluation:** Posted in the Course Orientation Module, along with the semester roadmap. Read both before the next lecture.
- **How this course runs:** You will be asked to commit to an answer before the instructor gives you one. Wrong first guesses are part of the method, not a failure of it.
- **One dataset, all semester:** You will carry a single dataset from first principles through to deployed models — meeting it as a business problem long before it becomes a matrix.

Why Data Mining Before Machine Learning

A fair question: this is a *machine learning* course, so why start with data mining?

> **DATA MINING ASKS:** What problems exist? Which questions are even worth asking of this data? What would count as a useful answer? Who acts on it? This is **problem formulation** — and it is where most real projects succeed or fail.

> **MACHINE LEARNING ASKS** how do we solve them? Given a well-posed problem and clean data, which algorithm minimizes which loss? This is a **method** — powerful, but useless if aimed at the wrong target.
>
> **THE HONEST REASON** An algorithm that answers the wrong question with 99% accuracy is worth **less than nothing** — it is confidently wrong, at scale. The first few lectures are about learning to recognize a good question, so that the algorithms you meet later have something worth pointing at.

ML is not being delayed. It is being aimed.

Where This Course Sits: A Short History

You are joining a conversation that is roughly sixty-five years old:

- **1960s — Statistics.** Inference from samples. Tukey argues that data analysis is a discipline in its own right.
- **1989 — Knowledge Discovery.** "KDD" was coined at IJCAI-89. The question becomes: what is worth finding? (DS605 starts here.)
- **1997 — Machine Learning.** Mitchell formalizes learning from experience. (The course arrives here later.)
- **2012 — Deep Learning.** AlexNet. Representations are learned, not hand-designed.
- **2022 — Foundation Models.** One pretrained model, many tasks.
- **2025 — Agentic AI.** Systems that act, not merely answer.

> **READ IT THE RIGHT WAY.** Each wave was built **on top of** the one before, not in place of it. Every later layer made answering easier — and none of them made the 1989 question go away: what is worth discovering, and who acts on it? That is still unanswered, and it is where the course begins.

The Recurring Framework

This five-stage strip reappears at the top of every major section, all semester:

- **Raw Data —** everything ShopKart records: orders, clicks, returns, payment events. Messy, incomplete, and far too large to read. (Today, and Lecture 02.)
- **Preprocessing —** cleaning, integrating, reducing, transforming. Unglamorous, and usually the majority of the real work. (Lectures 02–04.)
- **Pattern Discovery —** the six tasks introduced today, and later the ML algorithms that perform them at scale. (Today, then most of the semester.)
- **Knowledge —** a pattern that survived scrutiny: validated, generalizable, and understood well enough to explain to someone else. (Evaluation lectures.)
- **Decision —** someone changes what they do: review these 50 orders, call this customer, approve this loan. (Every case study, all semester.)

When you feel lost inside an algorithm later in the course, look up at this strip — it tells you which of the five problems you are actually solving.

Meet ShopKart: The Running Example

One company, one dataset, six different questions. Every example in this lecture draws on the same table, so you can concentrate on the ideas rather than re-learning a new industry every slide:

Customer	City	Tenure	Orders/mo	Avg basket	Payment	Last login	Returns
C-10241	Pune	26 mo	4.2	₹2,150	UPI	1 day ago	4%
C-10399	Jaipur	3 mo	1.1	₹740	CoD	29 days ago	31%
C-10402	Kochi	18 mo	3.7	₹1,890	Card	2 days ago	6%
C-10777 (flagged)	Delhi	1 mo	19.0	₹41,600	CoD	4 hrs ago	0%
C-10810	Surat	31 mo	2.9	₹1,320	UPI	6 days ago	9%

Five rows fit on a slide. ShopKart has **2.4 million**. That gap — between what a human can read and what an organization actually holds — is the entire reason this field exists.

Problem First: Spot the Pattern by Hand

Before any formal definition, the lecture asks students to look at six real ShopKart baskets and spot two products that seem to travel together:

Basket	Nappies	Wipes	Bread	Butter	Milk	Chips
#1	✔	✔	·	·	·	✔
#2	·	·	✔	✔	✔	·
#3	✔	✔	·	·	✔	·
#4	·	·	✔	✔	·	✔
#5	✔	✔	✔	·	·	·
#6	·	·	✔	✔	✔	·

Working it out by hand: Nappies and Wipes appear together in 3 of the 3 baskets containing Nappies — a 100% confidence pattern in this small sample, and the strongest relationship in the data. Bread and Butter also co-occur, but more weakly, and with only six baskets, it's impossible to tell whether that is a habit or a coincidence.

> **THE POINT OF THE EXERCISE:** You just did data mining by hand on six rows. The method you used in your head doesn't scale to 2.4 million rows. **That** is the problem the rest of the lecture formalizes.

The Definition of Data Mining

Data mining is the semi-automatic discovery of patterns in large datasets that are **implicit**, **non-trivial**, **previously unknown**, and **potentially useful** — drawing on machine learning, statistics, and database systems.

- **Implicit —** nobody wrote "beer goes with chips" into a column; you had to surface it.
- **Non-trivial —** not visible by skimming; it needed many records compared at once.
- **Previously unknown —** new to whoever is deciding; "obvious in hindsight" still counts.
- **Potentially useful —** someone can act on it; otherwise, it is a coincidence with good PR.

> **ALREADY, A WARNING:** Search hard enough through 2.4 million rows, and you **will** find striking patterns that mean nothing at all. Telling the difference is not a footnote to this field — it is the discipline of it.

Data Mining vs. Machine Learning

MACHINE LEARNING — THE ENGINE Algorithms that improve at a task through experience, without being explicitly reprogrammed (Mitchell, 1997). At ShopKart, the model that scores each order for fraud risk.

DATA MINING — THE WHOLE JOURNEY The goal-driven process of turning one real, messy dataset into knowledge someone acts on, using ML, statistics, and database methods together. At ShopKart: deciding which fraud cases to investigate, assembling the data, assessing whether the score is trustworthy, and setting the review threshold at 50.

IN INDUSTRY TODAY Teams rarely fail because their model was a few points less accurate. They fail because they optimized something the business never needed, or shipped a pattern nobody could act on. The engine was fine; the journey wasn't.

Think Like a Data Scientist: Baby Wipes and Churn

Scenario: ShopKart finds that customers who buy baby wipes are far more likely to cancel their subscription within 90 days. Should ShopKart stop selling baby wipes?

Things to consider before answering: Is the pattern real or an artifact of how you searched? What third factor might cause both? What would you need to measure to be sure?

CORRELATION IS NOT A MECHANISM. New parents buy wipes. New parents also have no time, less disposable income, and a collapsing attention span for shopping apps. The wipes did not cause the churn — **a life event caused both**. Ban the wipes, and you lose revenue while the churn continues, because you treated a symptom as a cause.

Good instinct to build now: whenever you see A → B, ask what C might be sitting behind both.

Recap: So Far, and What's Next

- **What we've established:** data mining exists to support decisions, it is a process rather than an algorithm, and patterns can mislead as easily as they inform.
- **The gap that opens up:** "find interesting patterns" is far too vague to act on. Interesting how? ShopKart has six different questions, and they are not the same shape.
- **What follows naturally:** the field's answer is a small, complete taxonomy of task types. Learn six shapes, and you can classify almost any problem you meet.

The Taxonomy: Six Questions for Any Dataset

Every one of these asks about the same ShopKart customers:

Predictive — fill in something unknown

- **Classification:** Will this customer churn? Yes or no?
- **Regression:** How much will they spend next quarter?

Descriptive — reveal existing structure

- **Cluster Analysis:** What kinds of customers do we even have?
- **Association Rules:** What gets bought together?
- **Sequential Patterns:** In what order do they buy it?
- **Anomaly Detection:** Which orders don't belong at all?

THE DISTINCTION THAT MATTERS Predictive tasks have a **right answer you can eventually check**. Descriptive tasks don't — nobody can tell you the "correct" segments, which makes them far easier to fool yourself with.

PART 2 · DESCRIPTIVE MINING

Nothing to predict. We're asking ShopKart's data to describe itself — and then asking whether we should believe it.

Cluster Analysis

Before the demo, the lecturer asks: "How many meaningfully different kinds of customers do you expect to find in 2.4 million ShopKart customers? The candidate answers and their honest assessments:

- **About 2 —** defensible; many businesses really do run on "high value" and "everyone else".
- **About 3 —** a common answer, and what the in-class demo produces. But notice: we are about to tell the algorithm to find three.
- **About 10 —** also defensible. Ten segments may be more truthful and completely unusable — marketing cannot run ten campaigns.
- **There's no true answer —** the most honest answer. There is no ground truth for segments. We choose a useful number, then argue for it.

The demo plots customers by order frequency and basket size (no labels exist) and runs k-means with k = 3. After the algorithm converges, three segments emerge purely from proximity — no labels were ever supplied. The open question the demo leaves you with: are these real marketing segments, or just three blobs? Run it again, and the boundaries shift.

Definition: Clustering partitions objects into groups so that members of a group are more similar to one another than to members of other groups, with no predefined labels.

WHEN IT FAILS — ASSUMPTIONS k-means assumes you already know k, that clusters are roughly round and similar-sized, and that "similar" means Euclidean distance. Break any of those, and it will still return an answer — confidently, and wrongly. It **never** reports "there are no groups here."

WHAT DECISION BECOMES POSSIBLE Three segments, three budgets: win back the lapsing, reward the loyal, nurture the new. Who benefits: marketing spends less for the same revenue; customers get fewer irrelevant messages.

BEYOND THE TEXTBOOK Foundation-model embeddings now let clustering run on text, images, and behavior in one shared space — so "similar customers" can be judged on what they wrote in a review, not just what they spent.

Stop and Think

The algorithm just drew three groups. Would you spend real money on them? Why?

Association Rules & Sequential Patterns

The "frequently bought together" strip is one of the highest-earning pieces of real estate in e-commerce, and underneath it is arithmetic simple enough to do by hand.

Association rules find {A} → {B} in baskets, scored by **support** (how often the pair occurs at all) and **confidence** (how often B appears given A). **Sequential patterns** add what baskets throw away: time.

Worked examples from the in-class basket demo (confidence/support):

Rule	Confidence	Support
Nappies → Wipes	82%	24%
Wipes → Nappies	79%	24%
Bread → Butter	85%	34%
Butter → Bread	80%	34%
Bread → Jam	60%	24%
Butter → Jam	55%	18%
Coffee → Sugar	75%	22%
Sugar → Coffee	70%	22%
Eggs → Milk	55%	18%
Nappies → Milk	44%	13%

THE CLASSIC TRAP High confidence ≠ , interesting. {anything} → {milk} looks strong simply because almost everyone buys milk. This is why **lift** exists — a 90%-confidence rule can still be worth nothing.

Anomaly Detection

Back to the 4,812 orders from the opening scenario. The demo plots last month's normal orders and lets you drop in tonight's suspicious order — remember customer C-10777: one month old, ₹41,600 basket, cash on delivery. The system learns a boundary from normal behavior (roughly 2.4 standard deviations from the mean, shown as a dashed circle) and flags points that fall outside it, reporting how many multiples of the boundary distance the new point is from the boundary.

THE DECISION This is how the 4,812 become a ranked shortlist of 50. Not a rule someone invented — a boundary *learned* from what normal has actually looked like.

WHERE THE FIELD IS MOVING Graph neural networks treat fraud as a network problem — catching rings of colluding accounts that each look perfectly normal in isolation.

Two important cautions surfaced from the demo itself: a flagged order is unusual, not fraudulent — a wedding order can look identical to a fraudulent attempt. And an order that looks ordinary and sails through is not proven safe — fraud that mimics normal behavior is exactly what a distance-based boundary will miss.

Think Like a Data Scientist: Is 98% Good?

Scenario: Your fraud model flags 50 orders a night with 98% accuracy. Fraud is about 1 in 500. Is 98% good?

Consider: what accuracy would a model that flags nothing achieve? Who pays the cost of a false positive, and are they in the room? Would your answer change in a hospital?

THE BASE-RATE TRAP A model that simply says "never fraud" scores **99.8%** — better than yours, while being completely useless. With rare events, accuracy is close to meaningless; you need precision, recall, and an explicit judgment about which error you would rather make. Every false positive is a real customer whose genuine order was frozen.

PART 3 · PREDICTIVE MINING

Now there is a specific unknown to fill in— and eventually reality will tell us whether we were right.

Classification (k-Nearest Neighbors)

The question: which ShopKart customers are about to leave? The full k-NN algorithm, in three lines:

1. Measure distance to every known customer: $d(x, x_i) = \sqrt{(\Sigma\square\ (x\square - x_i\square)^2}$
2. Keep the 3 smallest distances with k = 3.
3. Predict which label wins the vote among those neighbors.

In the demo, blue points are customers who stayed, and orange points are customers who churned. Clicking anywhere drops in a new customer whose fate is unknown, draws lines to its three nearest neighbors, and predicts by majority vote. A unanimous 3–0 vote is a confident region; a 2–1 split means the model is barely better than a coin toss near the boundary, yet it still reports an answer with the same apparent certainty.

COMMON BUSINESS OBJECTIVE Nobody wants a churn prediction for its own sake. They want a retention budget aimed at the **roughly 8% who are leaving and can still be persuaded**, which is a different, harder question.

Definition: Classification assigns a predefined categorical label to a new observation, learned from examples whose labels are already known.

ASSUMPTIONS & MISCONCEPTIONS It assumes tomorrow resembles yesterday — so a model trained before a festival sale, a competitor launch, or a pandemic quietly stops working. It also assumes your labels are true: if "churned" is defined as 90 days of silence, you have taught it a definition, not a reality.

WHAT DECISION BECOMES POSSIBLE? A ranked call-list for the retention team on Monday morning. Business value: retaining a customer typically costs a fraction of acquiring one, so even a mediocre model can pay for itself.

RESEARCH FRONTIER Causal ML asks not just who will churn, but which intervention would change it. Predicting churn perfectly while being unable to prevent any of it is a solved problem that helps nobody.

A Model Does Not Output a Decision

A prediction like "2–1 vote for churn" is not a fact — it is a probability wearing a confident face. The lecture walks through a chain: Prediction → Confidence → Decision → Outcome, using a slider from 1% to 99% confidence and a fixed action threshold of 0.50:

- **≥ 0.85 — confident, and rare.** Very few real customers ever look this clear-cut. A model that says 0.97 often is usually leakage, not skill.
- **0.55–0.85 — leaning churn, you act.** But roughly (1 − p) out of every 100 customers you call were never going to leave. That is the cost of acting.
- **0.45–0.55 — the model is essentially undecided.** Yet a decision still has to be made — this is the region where the threshold, not the model, is doing the work.
- **< 0.45 — leaning stay, you do nothing.** You quietly accept that about p/100 of these customers are walking out anyway. That is the cost of not acting.

The outcome of any single decision stays unknown for 90 days — the chain ends in genuine uncertainty, not a verified fact.

Stop and Think

Slide the cutoff up, and you call fewer people; slide it down, and you call more. Either way, some of those calls will be wrong. Which mistake would you rather make?

Regression

Same customers, different question: the answer is now a number, not a category. The demo plots ad spend against sales for past ShopKart marketing campaigns and fits a least-squares line to the data, reporting the equation sales ≈ m × spend + b along with R^2, and showing the residuals (the vertical gaps being minimized) in red.

> **READ THE LIMIT** The line happily predicts sales for ₹500 lakh of spend. No campaign in the data ever spent that. **Extrapolation is where regression lies most confidently.**

Failure Story: Google Flu Trends

In 2009, Google did something genuinely brilliant: it predicted flu outbreaks from search queries, faster than the CDC, at almost no cost. It was celebrated as the arrival of big data.

> **THEN IT QUIETLY STOPPED WORKING.** Between August 2011 and September 2013, it over-predicted flu in **100 of 108 weeks**. At the 2013 peak, it was estimated to be roughly twice the true rate — about 11% of Americans ill, compared with the CDC's 6%.

> **WHY** It never learned who was *ill* — it learned who was *searching*, and in a heavy flu news cycle, frightened healthy people search too. The correlation was real. The mechanism was never there.

> **WHY THIS BELONGS IN LECTURE ONE** The mathematics was not wrong. The engineering was not wrong. The data was enormous. It still failed — because nobody could say **why** the pattern should hold, so nobody noticed when the world moved, and the pattern didn't.

Source: Lazer, Kennedy, King & Vespignani, "The Parable of Google Flu: Traps in Big Data Analysis", Science 343(6176), 14 March 2014.

Stop and Think

Google Flu Trends would have passed every test run today, too. So what would make you go back and double-check a model that looks completely fine?

Synthesis: One Dataset, Six Lenses

These are not six topics — they are six questions you can put to the same ShopKart customers:

- **Classification — "Will they churn?"** Uses tenure, login recency, and order gaps to place each customer in a yes/no bucket. Decision: who the retention team calls on Monday.
- **Regression — "How much will they spend?"** Same columns, but the answer is a number, so success is measured in rupees of error rather than right-or-wrong. Decision: how much inventory to hold, and which customers justify a discount.
- **Clustering — "What kinds exist?"** No target at all; the structure comes purely from similarity. Decision: how many campaigns to run, and who receives each one.
- **Association — "what's bought together?"** Works on baskets rather than customers. Decision: bundles, shelf placement, and the "frequently bought together" strip.
- **Sequential — "in what order?"** Association plus time: phone → case → screen guard is a very different insight from all three in one basket. Decision: when to send the follow-up, not just what to recommend.
- **Anomaly — "What doesn't belong?"** Learns normal, then measures distance from it. Decision: which 50 of last night's 4,812 orders a human actually reads.

Quiz: Which Lens?

Four scenarios to test yourself before moving on — try to answer before reading the explanation.

1. "Which customers should we call before they leave?"

Options: Classification · Clustering · Association

Answer: Classification. A yes/no label you can check later. Note the business question is really "who can we still persuade?", which is subtly different.

2. "Group our customers however the data suggests — we have no idea what types exist."

Options: Classification · Cluster Analysis · Regression

Answer: Cluster Analysis. No labels, no target, just structure — and no way to be objectively "right."

3. "This ₹41,600 order from a one-month-old account doesn't look like anything we normally see."

Options: Anomaly Detection · Regression · Sequential Patterns

Answer: Anomaly Detection. Deviation from the learned normal. Unusual ≠ fraudulent, which is why a human still reviews it.

4. "How much revenue will the Diwali campaign bring in?"

Options: Classification · Regression · Association

Answer: Regression. A continuous number — and note it extrapolates beyond any spend level seen before, which is exactly where it is least trustworthy.

PART 4 · DATA MINING IN REALITY

Everything so far used a fictional company, so the ideas stayed clean. The obvious objection: *does any of this survive contact with a real business?* Two real companies, in the same industry as ShopKart, are examined next — deliberately, so you are comparing methods rather than adapting to a new domain.

Case Study: GoKwik — ShopKart's Problem, in Production

In Indian e-commerce, more than **30% of orders** are returned to origin (RTO) — shipped out, refused, shipped back. Cash-on-delivery orders are far worse than prepaid, and the loss is pure margin. Which lens is this? Classification, on exactly the ShopKart features already seen: tenure, payment method, and return history.

The decision enables scoring the risk at checkout, then selectively withdrawing the cash-on-delivery option for the riskiest buyers, before the parcel ever moves.

REPORTED OUTCOME: Around a **30% reduction in RTO**. Note what the model does not do: it never decides who is a fraudster — it changes which payment button renders.

NOTE THE ASYMMETRY A customer who wrongly scored high-risk silently loses a payment option they rely on, and is never told why.

Source: GoKwik product documentation; Entrepreneur India, June 2025.

Case Study: When the Data Itself Is the Attack

ShopKart's models trust their inputs. Real platforms can't: an estimated **42%** of reviews on major marketplaces may be fake or incentivized, and generative AI has made convincing ones nearly free.

The lens: classification again — genuine vs. fake — but now on text and reviewer behavior: posting times, incentive patterns, account history. Recent detectors report around **93%** accuracy on Amazon review data.

The twist worth remembering: Fakespot, the tool most textbooks cite as the pioneer here, was shut down by Mozilla on 1 July 2025. The problem did not go away; the solution did. Even case studies expire in this field.

THE ADVERSARIAL TURN Every other task in this lecture assumed data was merely messy. Here it is **actively hostile** — someone is being paid to defeat your model, and they update faster than your training set does. Almost nothing in a standard ML course prepares you for that.

Sources: FTC Consumer Reviews Rule (2024) · Amazon · TechXplore (2026) · Mozilla.

Think Like a Data Scientist: The Hospital Question

Scenario: A hospital asks you to deploy the exact anomaly detector you built for ShopKart — same maths — to flag patients whose vitals look unusual. What would you need before saying yes?

Trade-off to weigh: which is worse here, a missed emergency or fifty false alarms that exhaust the staff? Assumption to check: "unusual" was learned from whose bodies, and who is under-represented in that data? Accountability: when it's wrong at 3 a.m., who is answerable — the nurse, the hospital, or you?

THERE IS NO CLEAN ANSWER — THAT'S THE POINT. The mathematics transfers perfectly. The **consequences** do not. In retail, a false positive costs a customer an inconvenience. Here, it costs attention that another patient needed. Identical models carry entirely different obligations depending on where you point them — and deciding whether to deploy is part of your job, not someone else's.

The Data Scientist's Promise

Your job is NOT:

- to build models
- to maximize accuracy
- to create dashboards

Your job IS to help someone make a better decision, using incomplete, imperfect, uncertain data.

Everything else — the algorithms, the metrics, the code — is just how you keep that promise.

Recap Game: Predictive vs. Descriptive

Sort each of the six tasks into the correct family. Predictive tasks have a checkable right answer; descriptive ones don't:

- **Predictive:** Classification, Regression
- **Descriptive:** Cluster Analysis, Anomaly Detection, Association Rules, Sequential Patterns

Capstone Tease: One Dataset, the Whole Semester

ShopKart was a teaching device. Shortly, you will be given a real dataset — messier than anything on these slides — and you will keep it until the end of the course.

- **First, you'll interrogate it:** what questions can this data honestly answer? What is missing, wrong, or quietly biased?
- **Then, you'll prepare it:** clean, transform, and reduce it, and discover why practitioners say this is most of the work.
- **Finally, you'll model it:** build, evaluate, and defend real ML models on it, and be able to say why anyone should trust the result.

> **WHY ONE DATASET, NOT TWELVE TOY ONES?** Because the fifth time you return to the same data, you stop seeing rows and start seeing **the business behind them**. That shift is the actual learning outcome of this course.

PART 5 · CLOSING

A Question for Next Time

Today, you discovered patterns. Next lecture, we ask a harder question: what if the patterns themselves are lying to you?

ShopKart's customers mistype their city. A third of the return-reason fields are blank. Two systems disagree on what an "order" even is. Some rows were logged by bots. Google Flu Trends never saw it coming either.

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Closing

Questions? Bring the ones with no clean answer — those are the interesting ones.

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